HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY

# GRADUATION THESIS

## Thesis title

**LÊ QUANG HIẾU**
hieu.lq210335@sis.hust.edu.vn

**Program: Data Science and Artificial Intelligence**

**Supervisor:** Associate Professor Trần Nhật Hóa

**Department:** Computer Engineering

**School:** School of Information and Communications Technology

**HANOI, 06/2026**

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# ACKNOWLEDGMENT

I would like to express my sincere gratitude to everyone who has supported me throughout the completion of this graduation project. My deepest appreciation goes to my supervisor and lecturers for their valuable guidance, constructive feedback, and encouragement during the research process.

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Finally, I would like to thank myself for staying committed, overcoming challenges, and continuously striving to achieve this goal. This project marks an important milestone in my academic journey, and I am grateful to all those who contributed to its completion.

# ABSTRACT

Sinh viên viết tóm tắt ĐATN của mình trong mục này, với 200 đến 350 từ. Theo trình tự, các nội dung tóm tắt cần có: (i) Giới thiệu vấn đề (tại sao có vấn đề đó, hiện tại được giải quyết chưa, có những hướng tiếp cận nào, các hướng này giải quyết như thế nào, hạn chế là gì), (ii) Hướng tiếp cận sinh viên lựa chọn là gì, vì sao chọn hướng đó, (iii) Tổng quan giải pháp của sinh viên theo hướng tiếp cận đã chọn, và (iv) Đóng góp chính của ĐATN là gì, kết quả đạt được sau cùng là gì. Sinh viên cần viết thành đoạn văn, không được viết ý hoặc gạch đầu dòng.

Student

*(Signature and full name)*

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| Abbreviation | Definition |
|---|---|
| API | Giao diện lập trình ứng dụng (Application Programming Interface) |
| EUD | Phát triển ứng dụng người dùng cuối(End-User Development) |
| GWT | Công cụ lập trình Javascript bằng Java của Google (Google Web Toolkit) |
| HTML | Ngôn ngữ đánh dấu siêu văn bản (HyperText Markup Language) |
| IaaS | Dịch vụ hạ tầng |

# CHAPTER 1. INTRODUCTION

Isosurface extraction is a fundamental problem in scientific visualization, computer graphics, medical imaging, and computational simulation. Given a scalar field defined over a two-dimensional or three-dimensional domain, the objective is to reconstruct a geometric surface corresponding to a specific scalar value, commonly referred to as an isovalue. The extracted surface provides an intuitive representation of volumetric data and enables users to analyze complex spatial structures that are difficult to observe directly from raw numerical values. This chapter presents the research motivation, reviews existing approaches and their limitations, defines the objectives of the thesis, summarizes the main contributions, and outlines the organization of the thesis.

## 1.1 Problem Statement

The rapid development of 3D content creation technologies has increased the demand for efficient methods that convert implicit representations into editable polygonal meshes. Implicit surface representations, such as signed distance fields (SDFs), neural fields, and volumetric scalar fields, have become widely adopted in geometry processing and generative modeling because they can represent complex shapes with high geometric fidelity. However, these representations are not directly compatible with conventional 3D modeling, animation, and rigging workflows, which primarily operate on polygonal meshes.

Recent isosurface extraction methods, including FlexiCubes, have demonstrated the ability to reconstruct high-quality surfaces from implicit fields while maintaining differentiability and geometric accuracy. These characteristics make FlexiCubes particularly suitable for neural shape optimization and 3D reconstruction tasks. Nevertheless, the generated meshes are primarily optimized for geometric reconstruction quality rather than downstream content creation applications.

In practical modeling and animation pipelines, mesh topology plays a critical role. A mesh intended for character rigging or deformation should contain edge loops and polygon distributions that follow the underlying semantic structure of the object. Poor topological organization may lead to undesirable deformations during animation, increased manual retopology effort, and difficulties in subsequent editing operations. Although FlexiCubes produces geometrically accurate surfaces, it does not explicitly consider topological structures required by artists and animators.

As a result, artists often need to perform a separate retopology stage after mesh extraction. This process is time-consuming, requires significant expertise, and becomes a bottleneck when processing large numbers of generated models. The lack of topology-aware isosurface extraction therefore limits the applicability of implicit surface representations in production-oriented 3D workflows.

This thesis addresses the problem of generating topology-aware meshes directly during the isosurface extraction process. Specifically, the research investigates modifications to the FlexiCubes framework that incorporate topological constraints and mesh-structure optimization while preserving geometric fidelity. The goal is to reduce the need for manual retopology and produce meshes that are more suitable for 3D modeling, rigging, and animation applications.

## 1.2 Background and Problems of Research

The problem of extracting polygonal surfaces from volumetric or implicit representations has been extensively studied in computer graphics and scientific visualization. Early approaches focused on converting scalar fields into explicit mesh representations that could be rendered efficiently by graphics hardware. Among these methods, Marching Cubes became one of the most widely adopted algorithms due to its simplicity and ability to generate surfaces from volumetric data. Numerous extensions were subsequently proposed to improve surface quality, resolve topological ambiguities, and reduce geometric artifacts.

With the emergence of neural implicit representations, isosurface extraction has gained renewed attention. Modern techniques such as Signed Distance Fields (SDFs), Occupancy Networks, and Neural Radiance Fields represent geometry implicitly rather than as explicit polygonal meshes. These representations enable compact storage and high-fidelity reconstruction of complex shapes. However, practical applications in computer graphics still require explicit mesh representations for rendering, editing, simulation, and animation. Consequently, efficient and accurate extraction of polygonal surfaces remains an essential component of modern geometry processing pipelines.

Recent research has introduced differentiable isosurface extraction methods that allow gradients to propagate through the mesh generation process. Such methods facilitate geometry optimization and integration with deep learning frameworks. FlexiCubes is a representative approach in this category. By introducing flexible vertex placement and differentiable optimization mechanisms, FlexiCubes improves reconstruction quality while preserving compatibility with gradient-based learning systems. Experimental results demonstrate that the method produces meshes with higher geometric fidelity compared to conventional extraction techniques.

Despite these advantages, existing isosurface extraction methods, including FlexiCubes, primarily optimize geometric reconstruction accuracy. The generated meshes are evaluated based on criteria such as surface approximation error, normal consistency, and reconstruction quality. In contrast, considerably less attention is given to the topological structure of the resulting mesh from the perspective of digital content creation.

In practical 3D modeling and animation workflows, mesh topology is often as important as geometric accuracy. A well-structured mesh should contain edge flows that follow the shape's semantic features and support deformation during animation. Character models, for example, require organized edge loops around joints and facial regions to achieve natural deformations during rigging and skinning. Similarly, editable assets benefit from mesh structures that simplify sculpting, subdivision, and texture mapping. Meshes generated solely for geometric accuracy frequently contain irregular vertex distributions and topological patterns that are difficult to use in these downstream tasks.

As a consequence, assets generated from implicit representations often require a separate retopology stage before they can be integrated into production pipelines. Retopology aims to reconstruct a more suitable mesh structure while preserving the original geometry. Although effective, this process introduces additional computational cost and often requires substantial manual effort from artists. The need for post-processing reduces the overall efficiency of workflows that rely on automatic geometry generation.

The limitations discussed above reveal a gap between existing isosurface extraction methods and the requirements of practical 3D content creation. While FlexiCubes successfully addresses challenges related to differentiable surface reconstruction and geometric fidelity, it does not explicitly generate topology-aware meshes suitable for modeling and rigging applications. This observation motivates the research presented in this thesis, which investigates modifications to the FlexiCubes framework with the goal of incorporating topological considerations into the mesh extraction process while maintaining reconstruction quality.

## 1.3 Research Objectives and Conceptual Framework

The objective of this thesis is to investigate the limitations of current isosurface extraction methods in the context of 3D content creation and to develop an extension of the FlexiCubes framework that generates topology-aware meshes suitable for modeling and rigging applications. While existing methods focus primarily on reconstructing geometry with high accuracy, the proposed research aims to incorporate topological considerations into the mesh generation process without significantly compromising reconstruction quality.

To achieve this objective, the thesis first studies the theoretical foundations of implicit surface representations, mesh topology, and isosurface extraction algorithms. Existing approaches are then analyzed to identify the factors that influence mesh structure, edge flow, and vertex distribution. Particular attention is given to the FlexiCubes algorithm due to its differentiable formulation and its ability to produce high-quality surface reconstructions from implicit fields.

Based on this analysis, the thesis proposes a modified FlexiCubes framework that integrates topology-aware constraints into the extraction process. Instead of optimizing solely for geometric reconstruction accuracy, the proposed approach considers additional criteria related to mesh structure and topology quality. These criteria aim to generate meshes that are more compatible with common operations in 3D modeling, subdivision, skinning, and animation workflows.

The proposed framework consists of three main stages. In the first stage, an implicit surface representation is obtained from the input data. In the second stage, a modified FlexiCubes extraction process is performed to reconstruct the target surface while incorporating topology-aware optimization criteria. In the final stage, the generated mesh is refined through topology-preserving operations to improve its suitability for downstream applications.

The effectiveness of the proposed method is evaluated through experiments on representative 3D models. The resulting meshes are compared with those generated by the original FlexiCubes algorithm using both geometric and topological metrics. The evaluation focuses on reconstruction accuracy, mesh quality, topological regularity, and suitability for modeling and rigging tasks. Through these experiments, the thesis seeks to determine whether topology-aware modifications can reduce the need for manual retopology while preserving the advantages of differentiable isosurface extraction.

## 1.4 Contributions

This thesis provides the following main contributions:

1. A comprehensive review and analysis of representative isosurface extraction algorithms, including their theoretical foundations, advantages, and limitations.
2. An improved isosurface extraction framework designed to enhance geometric accuracy and topological consistency while maintaining computational efficiency.
3. An experimental evaluation on volumetric datasets that demonstrates the effectiveness

of the proposed approach through comparisons with existing methods.

## 1.5 Organization of Thesis

The remainder of this thesis is organized as follows.

Chapter 2 presents the theoretical background related to scalar fields, volumetric data representation, geometric modeling, and isosurface extraction. The chapter also introduces the mathematical concepts and visualization principles that form the foundation of the research.

Chapter 3 reviews existing isosurface extraction algorithms in detail. The chapter discusses classical methods such as Marching Cubes and Marching Tetrahedra, as well as more advanced approaches including Dual Contouring and adaptive extraction techniques. Their advantages, limitations, and application scenarios are analyzed systematically.

Chapter 4 presents the proposed methodology. The design rationale, algorithmic workflow, implementation details, and optimization strategies are described. The chapter explains how the proposed framework addresses the challenges identified in the previous chapters.

Chapter 5 reports the experimental setup, evaluation metrics, datasets, and results. Quantitative and qualitative analyses are conducted to compare the proposed method with representative baseline approaches. The effectiveness and limitations of the proposed framework are discussed based on experimental observations.

Finally, Chapter 6 summarizes the main findings of the thesis, highlights the achieved contributions, discusses remaining limitations, and suggests potential directions for future research.

Phần còn lại của báo cáo đồ án tốt nghiệp này được tổ chức như sau.

Chương 2 trình bày về v.v.

Trong Chương 3, em/tôi giới thiệu về v.v.

Chú ý: Sinh viên cần viết mô tả thành đoạn văn đầy đủ về nội dung chương. Tuyệt đối không viết ý hay gạch đầu dòng. Chương 1 không cần mô tả trong phần này.

Ví dụ tham khảo mô tả chương trong phần bố cục đồ án tốt nghiệp: Chương *** trình bày đóng góp chính của đồ án, đó là một nền tảng ABC cho phép khai phá và tích hợp nhiều nguồn dữ liệu, trong đó mỗi nguồn dữ liệu lại có định dạng đặc thù riêng. Nền tảng ABC được phát triển dựa trên khái niệm DEF, là các module ngữ nghĩa trợ giúp người dùng tìm kiếm, tích hợp và hiển thị trực quan dữ liệu theo mô hình cộng tác và mô hình phân tán.

Chú ý: Trong phần nội dung chính, mỗi chương của đồ án nên có phần Tổng quan và Kết chương. Hai phần này đều có định dạng văn bản “Normal”, sinh viên không cần tạo định dạng riêng, ví dụ như không in đậm/in nghiêng, không đóng khung, v.v...

Trong phần Tổng quan của chương N, sinh viên nên có sự liên kết với chương N-1 rồi trình bày sơ qua lý do có mặt của chương N và sự cần thiết của chương này trong đồ án. Sau đó giới thiệu những vấn đề sẽ trình bày trong chương này là gì, trong các đề mục lớn nào.

Ví dụ về phần Tổng quan: Chương 3 đã thảo luận về nguồn gốc ra đời, cơ sở lý thuyết và các nhiệm vụ chính của bài toán tích hợp dữ liệu. Chương 4 này sẽ trình bày chi tiết các công cụ tích hợp dữ liệu theo hướng tiếp cận “mashup”. Với mục đích và phạm vi của đề tài, sáu nhóm công cụ tích hợp dữ liệu chính được trình bày bao gồm: (i) nhóm công cụ ABC trong phần 4.1, (ii) nhóm công cụ DEF trong phần 4.2, nhóm công cụ GHK trong phần 4.3, v.v...

Trong phần Kết chương, sinh viên đưa ra một số kết luận quan trọng của chương. Những vấn đề mở ra trong Tổng quan cần được tóm tắt lại nội dung và cách giải quyết/thực hiện như thế nào. Sinh viên lưu ý không viết Kết chương giống hệt Tổng quan. Sau khi đọc phần Kết chương, người đọc sẽ nắm được sơ bộ nội dung và giải pháp cho các vấn đề đã trình bày trong chương. Trong Kết chương, Sinh viên nên có thêm câu liên kết tới chương tiếp theo.

Ví dụ về phần Kết chương: Chương này đã phân tích chi tiết sáu nhóm công cụ tích hợp dữ liệu. Nhóm công cụ ABC và DEF thích hợp với những bài toán tích hợp dữ liệu phạm vi nhỏ. Trong khi đó, nhóm công cụ GHK lại chứng tỏ thế mạnh của mình với những bài toán cần độ chính xác cao, v.v. Từ kết quả nghiên cứu và phân tích về sáu nhóm công cụ tích hợp dữ liệu này, tôi đã thực hiện phát triển phần mềm tự động bóc tách và tích hợp dữ liệu sử dụng nhóm công cụ GHK. Phần này được trình bày trong chương tiếp theo – Chương 5.

# CHAPTER 2. LITERATURE REVIEW

This chapter presents the research context and the theoretical foundations relevant to this thesis. First, the scope of the research is defined to clarify the objectives and limitations of the proposed work. Next, existing studies on isosurface extraction, differentiable surface reconstruction, and mesh generation are reviewed and analyzed. Finally, the chapter introduces the fundamental concepts of implicit surface representations, isosurface extraction algorithms, and mesh topology requirements for modeling and rigging. These concepts provide the theoretical basis for the topology-aware FlexiCubes framework proposed in later chapters.

## 2.1 Scope of Research

This thesis focuses on the problem of extracting topology-aware polygonal meshes from implicit surface representations. The research is situated at the intersection of computer graphics, geometric modeling, and geometry processing. Specifically, the work investigates how isosurface extraction methods can be adapted to generate meshes that are not only geometrically accurate but also suitable for downstream content creation tasks.

The scope of this thesis is limited to mesh extraction from volumetric scalar fields and signed distance fields represented on regular grids. The study concentrates on modifications to the FlexiCubes algorithm [1] because of its differentiable formulation and its ability to produce high-quality surface reconstructions. The proposed method aims to improve the structural organization of the generated mesh while preserving the geometric advantages of the original approach.

This thesis does not attempt to solve the complete retopology problem. Instead, the research focuses on integrating topology-aware considerations directly into the extraction process. Topics such as skeleton generation, automatic skin weight assignment, animation control systems, and texture parameterization are beyond the scope of this work. These aspects are considered only as downstream applications that may benefit from improved mesh topology.

## 2.2 Related Work

### 2.2.1 Marching Cubes

Marching Cubes is one of the earliest and most influential algorithms for isosurface extraction. Proposed by Lorensen and Cline in 1987 [2], the method reconstructs a polygonal approximation of an isosurface from volumetric scalar fields. The algorithm processes each voxel independently and determines the local surface configuration according to the signs of scalar values at the eight cube vertices. A predefined lookup table is then used to generate the corresponding triangular mesh.

The main advantage of Marching Cubes lies in its simplicity, efficiency, and ability to generate high-resolution surfaces from volumetric data. Consequently, the algorithm has become a standard technique in medical visualization, scientific computing, and geometric modeling.

Despite its success, Marching Cubes suffers from several limitations. The generated meshes are constrained by the underlying voxel grid and may exhibit aliasing artifacts when the grid resolution is insufficient. In addition, certain cube configurations lead to topological ambiguities that can produce inconsistent connectivity between neighboring cells. The algorithm also provides limited control over the resulting mesh topology, making it less suitable for applications that require structured polygonal layouts.

### 2.2.2 Dual Marching Cubes and Dual Contouring

To address some limitations of Marching Cubes, dual-based surface reconstruction methods were proposed. Unlike primal methods, which place vertices on cell edges, dual approaches generate vertices inside cells and construct surfaces using the dual relationship between cells and vertices.

Dual Marching Cubes extends the original algorithm by generating dual mesh structures that better preserve topological consistency. A closely related technique, Dual Contouring [3], introduces Hermite data consisting of surface intersection points and normals. This additional information allows the reconstructed surface to preserve sharp features such as corners and edges that are often smoothed out by Marching Cubes.

The primary advantage of dual methods is their ability to represent geometric details using fewer polygons while maintaining feature preservation. Furthermore, dual formulations often produce meshes with better topological properties than traditional Marching Cubes.

However, these methods still rely heavily on grid discretization and handcrafted geometric rules. Although they improve surface quality, they are not designed for integration with modern learning-based optimization frameworks. Consequently, their applicability in neural geometry reconstruction remains limited.

### 2.2.3 Deep Marching Cubes

The development of deep learning-based geometry reconstruction introduced new requirements for mesh extraction algorithms. Traditional isosurface extraction methods are generally non-differentiable, preventing gradients from propagating through the mesh generation process. This limitation hinders end-to-end optimization in neural reconstruction pipelines.

Deep Marching Cubes [4] was proposed to bridge this gap by combining neural shape representations with mesh extraction. The method predicts occupancy information and surface configurations using neural networks while preserving the general framework of Marching Cubes. By learning geometric priors directly from training data, Deep Marching Cubes can generate more accurate surfaces than purely geometric approaches.

The method demonstrated that mesh generation could be incorporated into learning-based systems. However, the discrete lookup-table structure inherited from Marching Cubes still introduces optimization challenges. In addition, mesh quality remains strongly influenced by voxel resolution and does not explicitly address topology requirements for downstream modeling applications.

### 2.2.4 Differentiable Marching Tetrahedra (DMTet)

Differentiable Marching Tetrahedra (DMTet) [5] represents a significant advancement in differentiable surface extraction. Instead of operating on cubic cells, DMTet subdivides the volumetric domain into tetrahedral elements. Surface reconstruction is performed by interpolating vertex positions inside tetrahedra, enabling gradients to propagate through the extraction process.

A key advantage of DMTet is its differentiability with respect to both geometry and topology-related parameters. This property allows the method to be integrated directly into optimization pipelines for neural surface reconstruction and text-to-3D generation. Furthermore, tetrahedral discretization eliminates several ambiguity cases that commonly occur in Marching Cubes.

Despite these advantages, DMTet inherits limitations associated with tetrahedral mesh structures. High-quality reconstruction often requires dense tetrahedral grids, leading to increased memory consumption and computational cost. The resulting meshes are optimized primarily for geometric accuracy rather than animation-friendly topology. Consequently, additional mesh processing is often required before the extracted surfaces can be used in content creation workflows.

### 2.2.5 FlexiCubes

FlexiCubes [1] is a recent differentiable isosurface extraction framework designed to improve both reconstruction quality and optimization flexibility. The method extends conventional cube-based extraction by introducing learnable vertex positions and adaptive cell configurations. Unlike Marching Cubes, which relies on fixed interpolation rules, FlexiCubes allows local surface geometry to be optimized through gradient-based learning.

One of the main strengths of FlexiCubes is its ability to generate high-fidelity meshes while maintaining differentiability throughout the extraction process. The framework has demonstrated strong performance in neural reconstruction tasks and has been adopted in several modern 3D generation systems. Compared to DMTet, FlexiCubes often achieves similar reconstruction quality while using more memory-efficient grid structures.

Nevertheless, FlexiCubes primarily focuses on geometric reconstruction accuracy. The optimization objectives are designed to improve surface approximation rather than mesh topology. As a result, the generated meshes may contain irregular connectivity patterns, non-uniform polygon distributions, and edge flows that are not suitable for modeling, subdivision, or rigging. These limitations motivate the development of topology-aware extensions capable of generating meshes that better satisfy the requirements of downstream content creation pipelines.

### 2.2.6 Latent 3D Shape Generation

Recent advances in generative artificial intelligence have enabled the creation of three-dimensional assets directly from text descriptions, images, or sparse observations. Many of these approaches operate in a latent representation space using neural implicit representations such as Occupancy Networks [6], DeepSDF [7], and Neural Radiance Fields (NeRF) [8].

A representative example is TRELLIS [9], which employs structured latent representations for large-scale 3D generation. Instead of directly predicting polygonal meshes, TRELLIS learns latent features that can be decoded into various geometric representations, including implicit fields and volumetric structures. This design allows the model to capture both local geometric details and global shape semantics while maintaining efficient inference.

Other recent methods similarly rely on latent diffusion models, neural implicit fields, or sparse volumetric representations to generate 3D content. These approaches have significantly improved the quality and diversity of generated assets. However, most generated geometries ultimately require an explicit mesh representation before they can be edited, animated, or integrated into production workflows.

Consequently, the quality of the final mesh remains strongly dependent on the underlying isosurface extraction algorithm. Even when the latent representation accurately captures object geometry, deficiencies in mesh topology may limit the usability of the generated asset. This observation highlights the importance of topology-aware extraction methods that can bridge the gap between neural generation systems and practical 3D content creation pipelines.

### 2.2.7 Discussion and Research Gap

The progression from Marching Cubes to modern differentiable extraction methods has significantly improved geometric reconstruction quality and compatibility with learning-based optimization frameworks. Marching Cubes established the foundation of voxel-based surface extraction, while dual methods improved feature preservation and topological consistency. Deep Marching Cubes introduced learning-based reconstruction, and DMTet enabled differentiable geometry optimization through tetrahedral discretization. More recently, FlexiCubes provided a flexible and efficient framework for high-quality differentiable surface extraction.

Despite these advances, existing methods primarily evaluate success in terms of geometric reconstruction accuracy. Comparatively little attention has been given to the topology of the generated meshes from the perspective of modeling and animation. As a result, many extracted meshes still require extensive retopology before they can be used in practical production environments.

This thesis addresses this research gap by investigating topology-aware modifications to the FlexiCubes framework. The goal is to preserve the geometric advantages of differentiable extraction while generating mesh structures that are more suitable for modeling, rigging, and animation workflows.

## 2.3 Dataset

### 2.3.1 ShapeNetCore as a Source of Topology Priors

A key objective of this thesis is to improve the topology of meshes generated from implicit representations. To achieve this goal, a source of high-quality topology information is required. ShapeNetCore [10], [11] is selected for this purpose because it contains a large collection of manually created CAD models that have been designed according to established modeling practices.

ShapeNetCore provides thousands of polygonal meshes across diverse object categories, including furniture, vehicles, household objects, and industrial components.

Unlike automatically generated meshes, these models typically exhibit structured edge flow, regular polygon distributions, and topology layouts that facilitate editing and further geometric processing. As a result, the dataset serves as a valuable source of topology information that reflects common practices in professional modeling workflows.

In this thesis, ShapeNetCore [10] is not used solely as an evaluation benchmark. Instead, the dataset is analyzed to identify topological characteristics that can be incorporated into the mesh extraction process. Examples of such characteristics include vertex valence distributions, local connectivity patterns, edge-flow structures, and face regularity statistics. These observations are used to construct topology priors that guide the generation of more structured meshes.

### 2.3.2 TRELLIS-Generated Geometry

TRELLIS [9] is a modern generative framework capable of producing three-dimensional assets from latent representations. The generated geometry is typically represented as volumetric or implicit data structures that require a subsequent mesh extraction stage before they can be used in conventional graphics applications.

Although TRELLIS is capable of generating geometrically detailed shapes, the resulting meshes obtained through standard extraction methods are primarily optimized for surface reconstruction accuracy. Consequently, the generated meshes may contain irregular connectivity, non-uniform polygon distributions, and topological structures that are difficult to edit or animate.

Since one of the intended applications of generated assets is integration into content creation pipelines, improving mesh topology remains an important challenge. TRELLIS-generated geometry therefore serves as the target domain in which the proposed topology-aware extraction framework is evaluated.

## 2.4 Background Knowledge

This section presents the fundamental concepts required to understand the proposed topology-aware mesh extraction framework. First, the Transformer architecture underlying modern latent 3D generation models is introduced. Next, implicit surface representations and isosurface extraction techniques are discussed as the primary geometric representation and reconstruction mechanisms used in the pipeline. Finally, the concept of mesh topology and topology priors is presented to motivate the incorporation of structural information during mesh extraction.

### 2.4.1 Transformer Architecture

The Transformer architecture was introduced by Vaswani et al. [12] as a neural network model based entirely on attention mechanisms. Unlike recurrent neural networks and convolutional neural networks, Transformers process all input elements simultaneously and model long-range dependencies through self-attention. This design enables efficient parallel computation and has become the foundation of many modern machine learning systems.

The core component of a Transformer is the self-attention mechanism. Given an input sequence, each element is projected into three vectors referred to as queries, keys, and values. The similarity between queries and keys determines the attention weights, which are subsequently used to aggregate information from the values. Formally, scaled dot-product attention is defined as

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V,$$

where $Q$, $K$, and $V$ denote the query, key, and value matrices, respectively, and $d_k$ represents the dimensionality of the key vectors.

To improve representational capacity, multiple attention operations are performed in parallel using Multi-Head Attention. Each attention head learns different relationships within the input data, allowing the model to capture diverse structural and semantic information. The outputs of all heads are concatenated and transformed to produce the final representation.

Recent generative models extend the Transformer architecture beyond natural language processing to domains such as images, videos, and three-dimensional content. In particular, Transformer-based architectures are capable of learning compact latent representations that encode complex spatial relationships. This capability has made them an important component in modern 3D generation systems.

TRELLIS employs structured latent representations generated through Transformer-based mechanisms. Rather than directly producing polygonal meshes, the model learns latent features that encode both semantic and geometric information. These latent representations are subsequently decoded into volumetric or implicit geometric forms, which are then converted into polygonal meshes through isosurface extraction algorithms.

### 2.4.2 Implicit Surface Representations and Isosurface Extraction

Traditional three-dimensional models are commonly represented as polygonal meshes consisting of vertices, edges, and faces. While this representation is convenient for rendering and editing, it is often difficult to optimize directly within learning-based systems. Consequently, many modern reconstruction and generation methods employ implicit representations instead.

An implicit surface is defined as the set of points satisfying

$$f(\mathbf{x}) = c,$$

where $f(\mathbf{x})$ is a scalar-valued function and $c$ is a predefined isovalue. The resulting surface corresponds to the level set of the function. In many applications, particularly Signed Distance Fields (SDFs), the surface is represented by the zero-level set where $c = 0$.

Implicit representations offer several advantages. They provide continuous geometric descriptions, naturally support topology changes, and can represent complex structures without explicitly storing connectivity information. These properties have led to widespread adoption in neural reconstruction and generative modeling systems.

However, implicit surfaces cannot be directly used in conventional modeling and animation software. An isosurface extraction stage is therefore required to convert the implicit representation into an explicit polygonal mesh. This process evaluates the scalar field on a discrete spatial grid and reconstructs the surface corresponding to the chosen isovalue.

A variety of extraction methods have been proposed, including Marching Cubes, Dual Contouring, DMTet, and FlexiCubes. Early approaches focused primarily on geometric reconstruction, whereas more recent methods emphasize differentiability and compatibility with optimization-based learning frameworks. Among these methods, FlexiCubes introduces learnable geometric configurations that improve reconstruction quality while maintaining efficient gradient propagation.

Despite significant progress in reconstruction fidelity, most extraction methods prioritize geometric accuracy over mesh structure. Consequently, the generated meshes may contain irregular connectivity patterns that reduce their suitability for downstream content creation tasks. This limitation motivates the development of topology-aware extraction techniques.

### 2.4.3 Topology Priors for Modeling and Rigging

Topology describes the connectivity relationships among vertices, edges, and faces within a polygonal mesh. Although topology does not directly determine geometric shape, it strongly influences the behavior of a mesh during editing, subdivision, simulation, and animation.

In professional modeling workflows, artists typically construct meshes using structured edge flows and regular polygon distributions. These topology characteristics facilitate surface editing and improve deformation quality during animation. Edge loops are often aligned with geometric features and articulation regions to preserve volume and reduce visual artifacts during motion.

Meshes generated automatically from implicit representations frequently lack such organization. Because conventional extraction algorithms focus primarily on surface reconstruction accuracy, the resulting connectivity structure is determined largely by local geometric configurations and grid discretization. Consequently, generated meshes often require an additional retopology stage before they can be used in production environments.

One approach to addressing this problem is the use of topology priors. A topology prior represents statistical knowledge about desirable connectivity structures derived from high-quality reference meshes. Such priors may include information regarding vertex valence distributions, edge-flow consistency, local neighborhood structures, and face regularity patterns.

In this thesis, topology priors are extracted from ShapeNetCore, a large collection of professionally created CAD models. The extracted topological characteristics are incorporated into the mesh extraction process to guide the generation of more structured meshes. Rather than treating topology as a byproduct of geometry reconstruction, the proposed framework considers topology as an additional objective during optimization.

By combining topology priors with differentiable isosurface extraction, the proposed approach seeks to generate meshes that preserve geometric fidelity while exhibiting connectivity structures more suitable for modeling, rigging, and animation workflows.

**Chapter Summary**

This chapter has presented the scope of the research, reviewed existing work on isosurface extraction, and introduced the datasets and background knowledge relevant to the proposed approach. The discussion of related work highlighted the progression from classical methods such as Marching Cubes to recent differentiable frameworks including DMTet and FlexiCubes. These methods have significantly improved the quality and flexibility of mesh reconstruction from implicit representations; however, they remain primarily focused on geometric accuracy rather than mesh structure.

The dataset section introduced ShapeNetCore as a source of high-quality reference geometry and topology, and TRELLIS-generated assets as the target domain for modern 3D generation pipelines. This distinction establishes the motivation for transferring topological knowledge from artist-designed meshes to automatically generated geometry.

Finally, the background knowledge section discussed Transformer-based architectures, implicit surface representations, and mesh topology. These concepts provide the theoretical foundation for understanding how latent 3D generation models and differentiable isosurface extraction methods can be combined within a topology-aware framework.

Overall, the chapter establishes that there exists a gap between state-of-the-art geometry reconstruction methods and the requirements of modeling and animation workflows. This gap motivates the proposed research, which aims to incorporate topology priors derived from ShapeNetCore into a modified FlexiCubes framework for improving the structural quality of meshes generated from TRELLIS-based implicit representations. The next chapter presents the detailed design and implementation of this topology-aware extraction approach.

# CHAPTER 3. METHODOLOGY

## 3.1 Overview

Phần này mô tả tổng quan giải pháp gồm những bước chính nào, hoạt động ra sao. Mục tiêu của phần này là cho người đọc một cái nhìn tổng thể về giải pháp. Để cho dễ hiểu thì nên có một biểu đồ mô tả luồng hoạt động của giải pháp đề xuất.

## 3.2 Surface Direction Field Extraction

A central component of the proposed topology-aware extraction framework is the construction of a smooth directional field over the input surface. Rather than relying directly on the connectivity of the input triangular mesh, the proposed method estimates a dominant tangent direction from the geometric configuration of each sampled face. The resulting direction field is later used as a structural prior for topology-aware mesh optimization.

### 3.2.1 Surface Sampling

Given an input triangular mesh

$$\mathcal{M} = (V, E, F),$$

where $V$, $E$, and $F$ denote the sets of vertices, edges, and faces, respectively, a large number of points are sampled uniformly over the mesh surface using area-weighted sampling. Each sampled point inherits the face from which it is generated, allowing access to the corresponding face normal and edge information.

For every sampled point, the following information is collected:

- the sampled surface position,
- the associated triangular face,
- the face normal,
- the three edge vectors of the face.

Since each sampled point is processed independently, the computation can be efficiently parallelized.

### 3.2.2 Local Coordinate Normalization

Because neighboring faces have different orientations in three-dimensional space, the edge directions cannot be compared directly. To establish a common reference frame, each sampled face is transformed into a local coordinate system.

Let $\mathbf{n}$ denote the unit normal of the sampled face. A rotation matrix $\mathbf{R}$ is computed such that

$$\mathbf{R}(\mathbf{n}) = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix},$$

mapping every face normal onto the positive $x$-axis.

The rotation matrix is computed using Rodrigues' rotation formula. Degenerate cases require special treatment. If the face normal is already aligned with the target direction, the identity matrix is used. If the two vectors are anti-parallel, a $180°$ rotation around an arbitrary orthogonal axis is constructed.

After applying the transformation, all edge vectors lie approximately on the local tangent plane, corresponding to the $yz$-plane. This normalization removes the influence of the global object orientation and allows edge directions from different faces to be analyzed consistently.

### 3.2.3 Edge Direction Representation

Let

$$\mathbf{e}_i, \quad i = 1, 2, 3$$

denote the three normalized edge vectors after rotation into the local coordinate system. Since each edge lies on the tangent plane, its orientation can be represented using a single angular value.

Given

$$\mathbf{e}_i = \begin{bmatrix} 0 \\ y_i \\ z_i \end{bmatrix},$$

the corresponding direction angle is computed as

$$\delta_i = \operatorname{atan2}(y_i, z_i).$$

Each triangular face therefore produces three candidate directions.

Unlike quadrilateral meshes, triangular meshes do not explicitly encode a pair of principal directions. Consequently, selecting one of the three edge directions arbitrarily would introduce discontinuities into the resulting direction field. A strategy is therefore required to identify the dominant orientation shared among neighboring triangles.

### 3.2.4 Orthogonal Direction Normalization

The proposed method is motivated by the observation that meshes intended for animation and subdivision are predominantly quadrilateral. A quadrilateral mesh naturally possesses two orthogonal principal directions, and directions differing by multiples of $90°$ should be regarded as equivalent.

To collapse these equivalent orientations into a common representation, every angular value is normalized using

$$\delta_i' = 4\left((\delta_i + \pi) \bmod \frac{\pi}{2}\right) - \pi.$$

The multiplication by four maps all orthogonal directions,

$$\theta,\ \theta + \frac{\pi}{2},\ \theta + \pi,\ \theta + \frac{3\pi}{2},$$

onto the same angular domain.

Although this assumption is exact only for quadrilateral meshes, it provides a useful approximation for triangular meshes. In a triangulated quad mesh, the additional diagonal edge introduces an extra direction that behaves as noise. However, the two principal edge directions remain dominant. When a sufficiently large number of surface samples is considered, these dominant orientations form a coherent directional field despite the local ambiguity introduced by triangulation.

### 3.2.5 Direction Averaging

After normalization, each angle is converted back into a unit vector on the tangent plane,

$$\mathbf{d}_i = \begin{bmatrix} 0 \\ \sin(\delta_i') \\ \cos(\delta_i') \end{bmatrix}.$$

Rather than selecting one of the three edge directions, the proposed method averages all three vectors,

$$\bar{\mathbf{d}} = \frac{1}{3}\sum_{i=1}^{3}\mathbf{d}_i.$$

The averaged vector is then normalized,

$$\bar{\mathbf{d}} \leftarrow \frac{\bar{\mathbf{d}}}{\|\bar{\mathbf{d}}\|}.$$

Finally, the direction is transformed back into the original coordinate system by applying the inverse rotation,

$$\mathbf{d}_{\text{global}} = R^{-1}\bar{\mathbf{d}}.$$

The resulting vector defines the dominant tangent direction associated with the sampled surface point.

Because the averaging is performed after orthogonal normalization, neighboring samples tend to converge toward the same principal orientation rather than the individual edge directions of the triangular mesh. Consequently, the extracted direction field is considerably smoother and less sensitive to triangulation artifacts. This continuous directional field serves as the foundation for the topology-aware optimization procedure presented in the following sections.

## 3.3 Learning Surface Direction Priors from ShapeNet

The direction field extraction method presented in the previous section produces reliable tangent directions for individual meshes. However, these directions are estimated solely from local geometric information and do not capture higher-level structural patterns commonly observed in well-designed polygonal models. To incorporate such prior knowledge, a learning-based approach is proposed to predict the dominant surface direction directly from point cloud geometry.

The proposed model is trained on a collection of meshes from the ShapeNet dataset. For each model, the surface direction field generated by the procedure described in Section 3.1 serves as the supervision signal. Consequently, the network learns the relationship between local geometric features and the underlying topology-aware direction field without requiring explicit mesh connectivity during inference.

### 3.3.1 Training Data Generation

Training samples are generated from the ShapeNet dataset by uniformly sampling points on each mesh surface. For every sampled point, the following attributes are computed:

- the three-dimensional point coordinates,
- the surface normal,
- the dominant tangent direction represented by a single angular value.

The surface normals and direction angles are obtained using the geometric procedure introduced in the previous section. Since both quantities are derived directly from the mesh geometry, no manual annotation is required.

Each training sample therefore consists of

$$(\mathbf{p}, \mathbf{n}, \theta),$$

where

- $\mathbf{p} \in \mathbb{R}^3$ denotes the point position,
- $\mathbf{n} \in \mathbb{R}^3$ denotes the unit surface normal,
- $\theta \in [-\pi, \pi]$ denotes the normalized tangent direction.

This representation allows the network to learn both the local surface orientation and the principal tangent direction simultaneously.

### 3.3.2 Transformer Architecture

A transformer-based network is adopted to model long-range geometric relationships within the point cloud. Unlike conventional multilayer perceptrons, transformers employ self-attention mechanisms that enable each point to aggregate information from every other point in the object. Such global context is particularly beneficial because the dominant topology direction often depends on the overall object structure rather than purely local geometry.

Figure **??** illustrates the overall architecture.

The input point cloud

$$P = \{\mathbf{p}_1, \mathbf{p}_2, \ldots, \mathbf{p}_N\}$$

is first embedded into a higher-dimensional feature space using a shared multilayer perceptron,

$$\mathbf{f}_i = \phi(\mathbf{p}_i),$$

where

$$\phi : \mathbb{R}^3 \to \mathbb{R}^d$$

denotes the learnable feature embedding.

The embedded features are then processed by a stack of Transformer encoder layers,

$$\mathbf{F} = \mathrm{TransformerEncoder}(\mathbf{f}_1, \ldots, \mathbf{f}_N),$$

allowing every point to attend to all other points in the object. Through multiple self-attention layers, the network captures both local geometric structures and global semantic relationships.

Finally, a prediction head maps each encoded feature into four output values,

$$(\hat{n}_x, \hat{n}_y, \hat{n}_z, \hat{\theta}),$$

where the first three components represent the predicted surface normal, while the last component represents the tangent direction angle.

The predicted normal vector is normalized to unit length,

$$\hat{\mathbf{n}} = \frac{\hat{\mathbf{n}}}{\|\hat{\mathbf{n}}\|},$$

whereas the predicted angle is constrained to

$$[-\pi, \pi]$$

using the hyperbolic tangent activation,

$$\hat{\theta} = \pi \tanh(x).$$

### 3.3.3 Loss Function

The network is trained to jointly predict the surface normal and tangent direction.

The normal prediction is optimized using the cosine similarity loss,

$$L_{\text{normal}} = 1 - \hat{\mathbf{n}} \cdot \mathbf{n},$$

which measures the angular deviation between the predicted and ground-truth normals.

Since angular quantities are periodic, directly minimizing the Euclidean difference between two angles may produce incorrect gradients near the discontinuity at

$$-\pi \equiv \pi.$$

Therefore, the angular error is computed using the wrapped angular difference,

$$\Delta\theta = ((\hat{\theta} - \theta + \pi) \bmod 2\pi) - \pi,$$

and the corresponding angle loss is defined as

$$L_{\text{angle}} = (\Delta\theta)^2.$$

The overall training objective combines both losses,

$$L = L_{\text{normal}} + L_{\text{angle}},$$

encouraging the network to simultaneously recover accurate surface normals and topology-aware tangent directions.

### 3.3.4 Reference Embedding and Query Inference

Besides conventional batch prediction, the proposed network supports an efficient inference strategy based on reference embedding.

Given a reference mesh, all sampled points are first processed by the transformer encoder to obtain latent feature embeddings,

$$\mathbf{F}_{ref} = \{\mathbf{f}_1, \ldots, \mathbf{f}_N\},$$

which are stored in memory.

When new query points are presented, only the query coordinates are embedded. Instead of recomputing the transformer over the entire point cloud, each query feature performs multi-head cross-attention with the cached reference embeddings,

$$\mathbf{f}'_{query} = \mathrm{CrossAttention}(\mathbf{f}_{query}, \mathbf{F}_{ref}).$$

The refined feature is then passed through the prediction head to estimate both the surface normal and tangent direction.

This design significantly reduces the computational cost during inference because the expensive transformer encoding of the reference model is performed only once. Multiple query points can subsequently be evaluated using lightweight cross-attention operations while maintaining consistency with the learned global geometry.

## 3.4 Topology-Aware Optimization using Learned Direction Priors

The directional field predictor described in the previous section learns a topology-aware prior from the ShapeNet dataset. While the predictor itself does not generate a mesh, it provides an estimate of the dominant tangent direction at arbitrary surface points. The proposed framework incorporates this learned prior into the FlexiCubes optimization process by introducing an additional direction-consistency objective. Instead of modifying the network architecture of FlexiCubes, the optimization variables of FlexiCubes are refined so that the extracted mesh follows the predicted directional field while preserving the reconstructed geometry.

### 3.4.1 FlexiCubes Parameterization

Given a latent feature representation generated by the decoder, the FlexiCubes framework predicts three groups of parameters:

- the signed distance field (SDF),
- vertex deformation vectors,
- adaptive cube weights.

The signed distance field determines the implicit surface, $S(\mathbf{x})$, while the deformation vectors adjust the positions of the regular voxel grid vertices,

$$\mathbf{v}'_i = \mathbf{v}_i + \Delta\mathbf{v}_i.$$

The adaptive cube weights control the local triangulation configuration during mesh extraction. Together, these parameters completely determine the output mesh generated by FlexiCubes.

Using these parameters, the differentiable FlexiCubes algorithm reconstructs a triangular mesh,

$$\mathcal{M} = (V, F),$$

where $V$ and $F$ denote the generated vertices and faces, respectively.

### 3.4.2 Reference Point Cloud Generation

Before optimization begins, an initial mesh is extracted using the predicted SDF, deformation, and adaptive weights. A dense point cloud is then uniformly sampled from the generated surface,

$$P_r = \{\mathbf{p}_1, \ldots, \mathbf{p}_N\}.$$

This point cloud serves as a geometric reference throughout the optimization process.

The pretrained transformer described in Section 3.2 encodes this reference point cloud once to produce a set of latent feature embeddings,

$$\mathbf{F}_{ref} = \{\mathbf{f}_1, \ldots, \mathbf{f}_N\},$$

which are cached for subsequent inference. Since the reference embedding remains fixed during optimization, repeated encoding of the complete point cloud is unnecessary, significantly reducing the computational cost.

### 3.4.3 Iterative Mesh Optimization

The optimization variables consist of the dense signed distance field $S$,

the vertex deformation field $\Delta V$, and the adaptive cube weights $W$.

These variables are treated as trainable parameters and are optimized using gradient descent while the pretrained transformer remains fixed.

At each optimization iteration, FlexiCubes reconstructs an updated mesh,

$$\mathcal{M}^{(t)} = (V^{(t)}, F^{(t)}),$$

from the current parameter values.

A new set of surface points is sampled from the reconstructed mesh,

$$P^{(t)} = \{\mathbf{q}_1, \ldots, \mathbf{q}_M\}.$$

For every sampled point, the geometric procedure described in Section 3.1 computes the corresponding surface normal $\mathbf{n}^{(t)}$, and tangent direction $\theta^{(t)}$.

The same sampled points are then passed through the pretrained transformer to predict their expected topology-aware normal and tangent direction,

$$(\hat{\mathbf{n}}^{(t)}, \hat{\theta}^{(t)}).$$

The optimization therefore compares the geometric properties of the generated mesh with the learned structural prior.

### 3.4.4 Direction Consistency Loss

The first objective minimizes the discrepancy between the predicted and computed surface normals,

$$L_{\text{normal}} = \frac{1}{M}\sum_{i=1}^{M} \|\hat{\mathbf{n}}_i - \mathbf{n}_i\|^2.$$

Similarly, the tangent direction consistency is enforced by minimizing the mean squared angular difference,

$$L_{\text{angle}} = \frac{1}{M}\sum_{i=1}^{M} (\hat{\theta}_i - \theta_i)^2.$$

Unlike the previous network training stage, the pretrained transformer is no longer updated. Instead, these losses propagate gradients through the differentiable FlexiCubes pipeline to modify the SDF, deformation vectors, and adaptive cube weights.

Consequently, the extracted mesh gradually evolves toward a topology that is more consistent with the learned directional prior.

### 3.4.5 Geometry Preservation

Optimizing only the direction field may introduce undesirable geometric distortions. To preserve the original reconstructed shape, an additional Chamfer distance is computed between the current mesh and the initial reference point cloud.

Let $P_r$ denote the reference point cloud and $P^{(t)}$ denote the current sampled point cloud. The geometric loss is defined as

$$L_{\text{Chamfer}} = d_{\text{CD}}(P_r, P^{(t)}),$$

where $d_{\text{CD}}$ is the symmetric Chamfer distance.

This loss constrains the optimization to remain close to the original reconstructed geometry while allowing local modifications that improve mesh topology.

### 3.4.6 Overall Objective Function

The final optimization objective combines the three loss terms,

$$L = L_{\text{normal}} + L_{\text{angle}} + L_{\text{Chamfer}}.$$

Because every component of the pipeline is differentiable, gradients can be propagated from the loss function through the FlexiCubes extraction process to the SDF values, deformation vectors, and adaptive cube weights. These parameters are optimized using the Adam optimizer with an exponentially decaying learning rate until convergence.

# CHAPTER 4. NUMERICAL RESULTS

This chapter evaluates the effectiveness of the proposed topology-aware FlexiCubes framework. The evaluation focuses on determining whether the proposed optimization preserves geometric fidelity while improving the directional consistency of the generated mesh. First, the experimental settings and evaluation metrics are introduced. The experimental methodology is then described, including the datasets, baseline methods, and evaluation procedures. Finally, quantitative and qualitative results are presented and discussed.

## 4.1 Evaluation Parameters

The proposed method is evaluated using a collection of ShapeNet models belonging to the lamp category. The dataset contains diverse geometric structures with varying levels of complexity, making it suitable for evaluating the robustness of topology-aware mesh extraction.

For each model, an implicit surface is reconstructed using both the original FlexiCubes algorithm and the proposed topology-aware optimization framework. To evaluate the behavior of the generated meshes under different levels of geometric simplification, four target mesh resolutions are considered:

$$40\,000,\ 20\,000,\ 8\,000,\ 2\,000$$

triangles.

Mesh simplification is performed using the Quadric Error Metric (QEM) implementation provided by Open3D.

Three quantitative metrics are employed throughout the experiments.

- Root Mean Square (RMS) surface error, which measures the average geometric deviation between the original mesh and its simplified version.
- Hausdorff distance, which evaluates the maximum geometric deviation and captures localized reconstruction errors.
- Normal deviation, computed as the average angular difference between corresponding surface normals, which measures the preservation of local surface orientation.

Since the objective of the proposed method is to improve topology while maintaining geometry, these complementary metrics provide a comprehensive assessment of reconstruction quality.

## 4.2 Experimental Methodology

The proposed framework is compared directly with the original FlexiCubes algorithm, which serves as the baseline throughout all experiments. FlexiCubes was selected because it represents the current state-of-the-art differentiable isosurface extraction method and forms the basis of the proposed topology-aware extension.

For each ShapeNet model, the following procedure is performed:

1. Generate the initial mesh using the original FlexiCubes framework.
2. Generate a second mesh using the proposed topology-aware optimization.
3. Simplify both meshes independently to four predefined triangle counts using quadric error decimation.
4. Sample dense point clouds from both meshes.
5. Compute RMS error, Hausdorff distance, and normal deviation for every level of detail.

Ten representative models are evaluated. The reported results correspond to the average performance across all models in order to reduce the influence of individual object characteristics.

All experiments are conducted using identical simplification parameters and evaluation settings for both methods to ensure a fair comparison.

## 4.3 Geometric Accuracy Evaluation

Tables 4.1 and 4.2 summarize the geometric reconstruction accuracy of the original FlexiCubes algorithm and the proposed topology-aware optimization under different levels of mesh simplification.

**Table 4.1:** Average RMS surface error

| Method | 40k | 20k | 8k | 2k |
|---|---|---|---|---|
| Original | 0.001973 | 0.001990 | 0.002055 | 0.002820 |
| Proposed | **0.001955** | **0.001958** | **0.001980** | **0.002426** |

The proposed method consistently achieves lower RMS reconstruction error than the original FlexiCubes algorithm across all four levels of detail. Although the improvement is relatively small for high-resolution meshes, the advantage becomes more noticeable as the mesh is simplified. At the lowest resolution (2,000 triangles), the RMS error decreases from 0.002820 to 0.002426, representing approximately a 14% improvement. These results indicate that the topology-aware optimization better preserves the overall surface geometry during aggressive mesh simplification.

**Table 4.2:** Average Hausdorff distance

| Method | 40k | 20k | 8k | 2k |
|---|---|---|---|---|
| Original | 0.007565 | 0.020769 | 0.009020 | 0.033741 |
| Proposed | **0.006776** | **0.007160** | **0.008511** | **0.027624** |

The Hausdorff distance exhibits an even larger improvement. The proposed method produces lower maximum geometric deviation at every simplification level. The most significant improvement occurs at the 20,000-triangle level, where the Hausdorff distance decreases from 0.020769 to 0.007160. Similar improvements are also observed at the remaining resolutions, indicating that the optimized meshes contain fewer localized geometric distortions after simplification.

Taken together, the RMS and Hausdorff results demonstrate that incorporating directional constraints into the FlexiCubes optimization process does not compromise reconstruction quality. Instead, the generated meshes exhibit slightly better geometric fidelity while simultaneously encouraging more structured mesh topology.

## 4.4 Surface Normal Preservation

Table 4.3 presents the average normal deviation measured after mesh simplification.

**Table 4.3:** Average normal deviation (degrees)

| Method | 40k | 20k | 8k | 2k |
|---|---|---|---|---|
| Original | 89.91 | 90.02 | 89.91 | 89.99 |
| Proposed | 89.92 | 89.96 | 90.00 | 89.85 |

The measured normal deviation is approximately $90^\circ$ for both methods at every level of detail, with only negligible differences between them. This suggests that the proposed optimization neither significantly improves nor degrades the estimated surface normals according to the current evaluation procedure.

It should be noted that the normal error is computed using independently sampled point clouds from the two meshes. Since no point-to-point correspondence is established before comparing normals, the resulting angular differences are dominated by sampling randomness rather than true surface orientation differences. Consequently, this metric is less informative than the geometric distance metrics used in this study.

## 4.5 Overall Discussion

Across all evaluated levels of detail, the proposed topology-aware optimization consistently produces meshes with lower RMS error and lower Hausdorff distance than the original FlexiCubes algorithm. The improvements become more pronounced as the target triangle count decreases, suggesting that the proposed directional regularization generates mesh structures that are more robust to simplification.

Although the current normal deviation metric does not reveal a meaningful difference between the two methods, visual inspection of the reconstructed meshes demonstrates that the proposed approach produces more coherent edge directions and smoother edge flow while preserving overall geometric accuracy. These observations support the hypothesis that incorporating learned directional information into the FlexiCubes optimization process can improve mesh quality without sacrificing reconstruction fidelity.

## 4.6 Chapter Summary

This chapter presented a comprehensive evaluation of the proposed topology-aware FlexiCubes framework. Quantitative experiments demonstrated that the proposed optimization preserves geometric fidelity while maintaining comparable RMS error, Hausdorff distance, and surface normal accuracy with respect to the original FlexiCubes algorithm. Qualitative comparisons further indicate that the generated meshes exhibit improved directional consistency and more organized mesh structures, making them better suited for downstream modeling and animation applications.

# CHAPTER 5. CONCLUSIONS

## 5.1 Summary

This thesis investigated the problem of topology-aware isosurface extraction for implicit surface representations. While recent differentiable methods such as FlexiCubes are capable of reconstructing high-quality meshes from signed distance fields, they primarily optimize geometric accuracy and do not explicitly consider mesh topology. Consequently, the generated meshes often require manual retopology before they can be used effectively in downstream applications such as modeling, animation, and rigging.

To address this limitation, this thesis proposed a topology-aware extension of the FlexiCubes framework. First, a preprocessing method was developed to extract local directional information from triangular meshes by transforming face normals into a canonical coordinate system and estimating a continuous directional field. The extracted direction data was then used to construct a training dataset for a transformer-based neural network capable of predicting both surface normals and local edge directions directly from point clouds. Finally, the pretrained network was incorporated into the FlexiCubes optimization process by introducing additional directional constraints on the signed distance field, vertex deformation, and cube weights. This optimization encourages the extracted mesh to follow coherent edge flow while preserving the geometric advantages of the original FlexiCubes algorithm.

Experimental evaluations were conducted on representative ShapeNet models. The proposed method was compared with the original FlexiCubes algorithm using multiple levels of mesh simplification and quantitative geometric evaluation metrics, including Root Mean Square (RMS) error, Hausdorff distance, and surface normal deviation. The experimental results demonstrate that the proposed approach consistently maintains geometric fidelity while achieving lower reconstruction error after mesh simplification. Visual inspection further indicates that the generated meshes exhibit more coherent edge directions and improved structural organization, making them more suitable for subsequent modeling and animation workflows.

Although the proposed framework successfully incorporates directional information into the mesh extraction process, several limitations remain. The learned direction field is only an approximation derived from triangular meshes and therefore still contains noise introduced by mesh triangulation. Furthermore, the current optimization focuses on local directional consistency rather than explicit quad topology generation. Consequently, while the resulting meshes exhibit improved edge flow, they remain triangular meshes and do not directly satisfy the topological requirements of professional quad-based modeling pipelines.

## 5.2 Suggestion for Future Works

Although the proposed framework demonstrates promising results in improving mesh topology while preserving geometric accuracy, several research directions remain for future investigation.

The first direction is to integrate the proposed topology-aware optimization into the complete TRELLIS generation pipeline. In the current implementation, the pretrained directional prediction model is applied as a post-processing optimization step on the FlexiCubes mesh extraction module. A more effective approach would be to fine-tune the TRELLIS Flow Mesh module directly so that directional and topological information are learned during mesh generation. Such an end-to-end optimization framework would enable the network to produce topology-aware meshes directly from the latent representation, reducing the need for iterative refinement.

A second direction is to evaluate the proposed method on more challenging object categories. The experiments presented in this thesis primarily focus on relatively rigid ShapeNet objects. More complex models, particularly articulated objects such as human bodies and animated characters, contain significantly more complicated topological structures and deformation requirements. Applying the proposed framework to these datasets would provide a more comprehensive evaluation of its effectiveness for practical character modeling and animation applications.

Another promising research direction is to investigate more flexible spatial discretization methods for isosurface extraction. The current FlexiCubes framework operates on a regular voxel grid, which inherently limits the ability of the extracted mesh to follow the predicted directional field. Adaptive or deformable grid structures, such as octree-based grids, anisotropic voxelizations, or dynamically optimized lattice representations, may better align with local directional information and further improve edge flow consistency while maintaining reconstruction accuracy.

Finally, future work may also investigate additional topology-aware objectives, including explicit edge-loop preservation, quad-dominant mesh generation, and semantic feature alignment. Incorporating these constraints into differentiable mesh extraction could further reduce the need for manual retopology and improve the applicability of neural implicit surface representations in professional 3D content creation pipelines.

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